

Lyapunov Functions: An Optimization Theory Perspective[★]

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Abstract: The problems of unconstrained optimization and establishing asymptotic stability have much in common. Understanding the analogy between these two sheds light on their interconnection and may lead to a number of new results. For instance, in this paper, we provide estimates of the rate of convergence when analyzing asymptotic stability of differential equations, rather than just ascertain the very fact of stability. Also, standard methods for the design of Lyapunov functions (e.g., those having the meaning of the full energy of the system) turn out to be unsatisfactory from this point of view and have to be modified. These claims are exemplified in the paper by considering the heavy-ball method for function minimization “in parallel” with the problem of asymptotic stability for the synchronous motor equation.

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1. INTRODUCTION

At first glance, the problems of unconstrained optimization and asymptotic stability represent quite separate fields of research. However they have much in common, both in the problem formulation and the techniques exploited. The unconstrained minimization problem is formulated as follows: Given a function to be minimized, design an algorithm (in the form of either difference or differential equation) and prove its convergence to a minimum point. Conversely, the analysis of asymptotic stability of an equilibrium point of a difference or differential equation is traditionally performed via use of Lyapunov’s direct method. Namely, a Lyapunov function is constructed such that it decreases monotonically along the trajectories of the system. In other words, the given data and the design goals in both problems change places. We summarize this concept in the table below:

Problem	Optimization	Stability
Data	Function	Equation
Technique	Design a method	Construct a Lyapunov function
Goal	Prove convergence	Prove stability

These links between the two fields enable their mutual enrichment both in terms of techniques and results. First, in the optimization theory, one can exploit the diversity of Lyapunov functions beyond the standard candidates (objective function or the distance to a minimum point). We

will demonstrate this possibility for the heavy-ball method in unconstrained minimization. On the other hand, in the asymptotic stability analysis, it was traditional just to prove stability, while the rate of convergence was of a lesser interest; moreover, it cannot be assessed with the commonly used technique based on LaSalle’s invariance principle. Needless to say that in optimization, finding such estimates is the key issue. Also, in optimization problems, it is often the case that the minimum point is not unique or may not exist. Similar nonstandard assumptions, the non-uniqueness of the equilibrium, are of interest in the analysis of difference and differential equations.

More precisely, the content of the paper is as follows. As it is well known, the concept of Lyapunov function is the basic tool in the analysis of asymptotic stability for ODE. The idea of Lyapunov’s direct method was formulated in the fundamental work (Lyapunov, 1892); at present, there exist dozens of monographs and textbooks on the subject; e.g., see the classical works (Andronov et al., 1966; LaSalle and Lefschetz, 1961) and more recent publications (Halanay and Răsvan, 1993; Haddad and Chellaboina, 2008; Malisoff and Mazenc, 2009; Giesl and Hafstein, 2015). Other modern results in this field such as (Angeli, 2004; Dashkovskiy et al., 2011; Angeli and Efimov, 2005) exploit the fruitful concept of the input-to-state stability proposed in (Sontag, 1995); they are also based on the Lyapunov function technique. In Section 2 we present the main theorems on Lyapunov’s direct method under nonstandard assumptions. For instance, we address situations with non-

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unique equilibrium points or the absence of equilibria. We focus on the estimates of the rate of convergence; sometimes the estimates are obtained for the averaged process, not the original one.

In Section 3, two basic continuous-time methods for unconstrained minimization, the gradient descent and the heavy-ball method are considered, see (Polyak, 1964; Attouch et al., 2000; Brown and Bartholomew-Biggs, 1989; Begout et al., 2015). We prove that both methods guarantee the convergence of the gradient to zero under minimal assumptions. Under slightly stronger assumptions we prove convergence of the function values to the minimum and estimate the rate of convergence. Highly importantly, these results are global and they do not require convexity. We also demonstrate the variety of Lyapunov functions suitable for establishing the convergence.

In Section 4 we analyze one of the classical pendulum-like equations, forced torque pendulum, or equation for synchronous motor (Tricomi, 1933; Andronov et al., 1966; Leonov et al., 1992). This equation is rich enough to exhibit the drawbacks of the classical energy-like Lyapunov functions and to design new families of such functions. A multidimensional extension of this problem (swing equation, dynamic model of power systems) is of great interest (Pai, 1989; Vu and Turitsyn, 2016; Machowski et al., 2012; Weitenberg et al., 2016; Zao et al., 2015). Actually, it is the applications to power systems that was one of the main motivations for the present research.

Finally, Section 5 summarizes the results obtained and indicates possible extensions of the approach.

2. LYAPUNOV'S DIRECT METHOD

Various parallels between this method and algorithms of unconstrained optimization have been drawn many times; e.g., see (Polyak, 1987; Brown and Bartholomew-Biggs, 1989). We present several basic assertions in the form convenient for the exposition to follow; these will be formulated using the notation of the optimization theory, which is somewhat different from the one usually adopted in the theory of asymptotic stability.

We consider the autonomous (time-invariant) nonlinear differential equation

$$\dot{x} = \phi(x), \quad x(0) \in \mathbb{R}^n, \quad (1)$$

and a continuously differentiable function $V(x)$, which is bounded from below, $V(x) \geq v^*$. Denote by

$$\dot{V}(x) = (V'(x), \phi(x))$$

its time derivative along the solutions of (1), where the sign ' stands for the gradient. The theorem below is a summary of known results on Lyapunov's method, which are, however, presented in a slightly modified form motivated by optimization applications.

Theorem 1. Assume that $V(x)$ is bounded from below, $V(x) \geq v^*$, and there exists a continuous $W(x)$ such that $-\dot{V}(x) \geq W(x) \geq 0$ for all x . Then the following statements hold true:

1. For the solutions of (1) we have $\liminf_{t \rightarrow \infty} W(x(t)) = 0$ as $t \rightarrow \infty$ and

$$\min_{0 \leq t \leq T} W(x(t)) \leq \frac{V(x(0)) - v^*}{T}.$$

2. Assume $W(x)$ is convex and denote $\bar{x}(T) = \frac{1}{T} \int_0^T x(t) dt$.

Then

$$W(\bar{x}(T)) \leq \frac{V(x(0)) - v^*}{T}.$$

3. Assume that

- (i) $V(x^*) = v^*$ with $V(x) > v^*$ for $x \neq x^*$,
- (ii) $\inf_{\|x - x^*\| \geq R} (V(x) - v^*) \geq \varepsilon > 0$ for $R > 0$, and
- (iii) $W(x^*) = 0$ with $W(x) > 0$ for $x \neq x^*$.

Then $x(t) \rightarrow x^*$ as $t \rightarrow \infty$.

4. If $W(x) \geq \gamma(V(x) - v^*)$, $\gamma > 0$, then

$$V(x(t)) - v^* \leq (V(x(0)) - v^*) \exp(-\gamma t).$$

Note that in statements 1, 2, we neither assume the existence of equilibria points for equation (1) nor claim the convergence of $V(x(t))$ to zero. This situation is typical to optimization problems, where the existence of the minimum point sometimes cannot be ascertained *a priori*, or convergence to this point cannot be guaranteed. Instead we provide some estimates for the decay rate of $W(x(t))$. Note also that we do not use the standard terminology of the stability theory (such as: “ $V(x)$ is positive definite if and only if $V(x^*) = 0$, $V(x) > 0$ for $x \neq x^*$; and $V(x)$ is positive semidefinite if and only if $V(x^*) \geq 0$ for all x ”), because it is in some contradiction with the terminology in optimization.

Proof. Proof of statement 1 is immediate; indeed, integrating the inequality $-\dot{V}(x(t)) \geq W(x(t))$ from 0 to T we obtain

$$\int_0^T W(x(t)) dt \leq V(x(0)) - V(x(T)) \leq V(x(0)) - v^*.$$

If W is convex, Jensen's inequality implies statement 2. Statement 3 is a classical form of the Lyapunov theorem on asymptotic stability. Finally, applying the Grönwall–Bellman inequality to $-\dot{V} \geq \gamma(V(x) - v^*)$ we arrive at statement 4.

Theorem 1 presents the global convergence results and global asymptotic stability property. The result below provides the simplest local version of this theorem.

Theorem 2. Let the assumptions of Theorem 1 be satisfied in some bounded nonempty domain

$$D = \{x : V(x) \leq h, x \in Q\},$$

where Q has nonempty interior Q^0 , and $D \subset Q^0$. Then the statements of Theorem 1 remain valid for $x(0) \in D$.

Indeed, the assumptions $V(x(0)) \leq h$ and $W(x) \geq 0$ guarantee that $x(t) \in D$ for all $t \geq 0$. Note that $V(x)$ is not assumed to be bounded from below on the entire space $\mathbb{R}^n$; we just require the boundedness of the component of the level set $V(x) \leq h$ in Q . This situation will be met in Section 4.

3. GRADIENT METHOD AND HEAVY-BALL METHOD

Consider the unconstrained minimization problem

$$\min f(x), \quad x \in \mathbb{R}^n. \quad (2)$$

We assume that $f(x) \in C^1$; then the simplest minimization method is the gradient descent

$$\dot{x} = -f'(x), \quad (3)$$

which dates back to Cauchy. The results on its convergence can be found in (Polyak, 1963) and more recent publications. A number of results can be derived from Theorem 1.

Theorem 3. The following assertions hold true:

1. If $f(x)$ is bounded from below, $f(x) \geq f^*$, then

$$\min_{0 \leq t \leq T} \|f'(x(t))\|^2 \leq \frac{f(x(0)) - f^*}{T}.$$

2. If, in addition

$$\|f'(x)\|^2 \geq 2l(f(x) - f^*), \quad l > 0, \quad (4)$$

we have

$$f(x(t)) - f^* \leq (f(x(0)) - f^*) \exp(-2lt).$$

3. If $f(x)$ is convex and there exists x^* such that $f(x^*) = f^*$, then

$$f(\bar{x}(T)) - f^* \leq \frac{\|x(0) - x^*\|^2}{2T}.$$

4. If $f(x)$ is strongly convex¹, then $x(t) \rightarrow x^*$ exponentially as $t \rightarrow \infty$.

Proof. To prove the first two assertions, the function V can be taken in the form $V(x) = f(x)$, in which case we have $W(x) = \|f'(x)\|^2$. Then statements 1 and 4 of Theorem 1 imply the desired result.

For the last two assertions, adopting $V(x) = \frac{1}{2}\|x - x^*\|^2$ leads to $W(x) = f(x) - f^*$. Now statement 2 of Theorem 1 implies assertion 3, while for strongly convex functions, the minimum point x^* exists and inequality (4) is satisfied, hence, statement 4 of Theorem 1 implies assertion 4.

Note that condition (4) does not assume neither the convexity of $f(x)$ nor the uniqueness of the minimum point. For instance, if $g: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $f(x) = \|g(x)\|^2$, and $g'(x)$ has rank m for all x , then $f(x)$ is nonconvex (for nonlinear $g(x)$), but (4) is satisfied; see (Nesterov and Polyak, 2006), where this inequality (and its extensions) is called the *gradient dominance* condition. On the other hand, for $f(x) = \|Ax - b\|^2$, the condition holds true even if the smallest singular value of the matrix A is zero.

We now turn to the analysis of the heavy-ball method, which has the following form:

$$\ddot{x} + a\dot{x} + bf'(x) = 0; \quad (5)$$

it was derived in (Polyak, 1964) from physical analogies with the motion of a body in the potential field $f(\cdot)$ subject to friction. Here, $x(t) \in \mathbb{R}^n$ is the trajectory of the body, $\dot{x}, \ddot{x}$ are its velocity and acceleration, and $a, b > 0$ are scalar parameters. It is natural to assume that, as time evolves, the body reaches a local minimum of the potential $f(x)$; i.e., this methods solves the minimization problem. In (Polyak, 1964, 1987), results on the local convergence and the rate of convergence in (5) were obtained using the linearization techniques; global convergence was proved for quadratic functions $f(x)$. Certain results for the

non-quadratic case can be found in (Attouch et al., 2000; Brown and Bartholomew-Biggs, 1989; Begout et al., 2015); below we present some modifications of these results.

First, we re-write equation (5) in the form (1):

$$\begin{aligned} \dot{x} &= y, \\ \dot{y} &= -ay - bf'(x). \end{aligned}$$

There is much freedom in choosing the form of the function $V(x, y)$. From mechanical analogies, it seems very reasonable to adopt this function in the form of the full energy of the system (the sum of the potential and kinetic energy) as $V = f(x) + \frac{1}{2b}\|y\|^2$. For the time derivative over the solutions of the system we then have

$$\dot{V}(x, y) = (f'(x), y) + \frac{1}{2b}2(y, -ay - bf'(x)) = -\frac{a}{b}\|y\|^2,$$

so that $W = \frac{a}{b}\|y\|^2 \geq 0$, and asymptotic stability can be established via use of LaSalle's invariance principle (known in the Russian literature as the Barbashin–Krasovskii theorem). However, any stronger assertions as well as estimates of the rate of convergence are impossible to obtain on this way.

On the other hand, recall that, even for linear systems, a desired quadratic Lyapunov function (Px, x) for the system involves the matrix P having nonzero off-diagonal entries. In other words, we may try to construct a strictly decreasing Lyapunov function by adding a cross-term to V ; e.g., see (Zao et al., 2015). Namely, consider the function of the form

$$V = (f(x) - f^*) + \frac{a}{a^2 + 2bL}(f'(x), y) + \frac{L}{a^2 + 2bL}\|y\|^2, \quad (6)$$

where f^* is a lower bound on $f(x)$ and L is an upper bound on $\|f''(x)\|$. Then we are in a position to apply Theorem 1, and the following result holds.

Theorem 4. Let $f(x)$ be twice differentiable with $\|f''(x)\| \leq L$, $f(x) \geq f^*$, and $a > 0$, $b > 0$. Then for any $x(0)$, $y(0)$, for method (5) we have

$$\min_{0 \leq t \leq T} \|f'(x(t))\|^2 \leq \frac{V(0) - f^*}{cT}, \quad c = \frac{ab}{a^2 + 2bL}. \quad (7)$$

If, in addition, condition (4) holds, then the convergence $f(x(t)) \rightarrow f^*$ is exponential.

Proof. First, for functions with Lipschitz gradient we have $f(x) - f^* \geq \frac{1}{2L}\|f'(x)\|^2$, hence

$$\begin{aligned} V &\geq \frac{1}{2L}\|f'(x)\|^2 + \frac{a}{a^2 + 2bL}(f'(x), y) + \frac{L}{a^2 + 2bL}\|y\|^2 \\ &\geq c_1(\|f'(x)\|^2 + \|y\|^2), \quad c_1 > 0. \end{aligned}$$

Then, computing the derivative $\dot{V}$, we obtain

$$\dot{V} \leq -\frac{La}{a^2 + 2bL}\|y\|^2 - \frac{ab}{a^2 + 2bL}\|f'(x)\|^2,$$

which implies the estimate

$$W \geq c_2(\|f'(x)\|^2 + \|y\|^2), \quad c_2 > 0.$$

Therefore, statement 1 of Theorem 1 applies, and we arrive at (7). Use of statement 4 of Theorem 1 is valid by condition (4).

For convex $f(x)$, different Lyapunov functions can be exploited. Borrowing the idea of (Su et al., 2016), consider the following Lyapunov function:

¹ A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be strongly convex with constant $\ell > 0$ if $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y) - \frac{\ell\lambda(1 - \lambda)}{2}\|x - y\|^2$ for all $x \neq y$ and $0 \leq \lambda \leq 1$, see (Polyak, 1987).

$$V = (f(x) - f^*) + \frac{a^2}{2b} \|x - x^* + \frac{1}{a}y\|^2, \quad (8)$$

where x^* is a minimizer of $f(x)$.

Theorem 5. If $f(x)$ is convex and $f(x^*) = f^*$, then for method (5) and $\bar{x}(T) = \frac{1}{T} \int_0^T x(t) dt$ we have

$$f(\bar{x}(T)) - f^* \leq \frac{V(x(0), y(0))}{aT}. \quad (9)$$

Proof. Computing the derivatives, we find

$$\dot{V} = -a(f'(x), x - x^*),$$

so that $W(x)$ can be taken as $W(x) = a(f(x) - f^*)$. Estimate (9) then follows by applying statement 2 of Theorem 1.

In (Su et al., 2016), the authors considered method (5) with the t -dependent parameter a , $a = c/t$. They used the Lyapunov function of the sort (8) with time-dependent parameters and proved the convergence with rate $O(1/t^2)$. The same paper contains numerical comparison of various versions of gradient, heavy ball and similar methods.

4. SYNCHRONOUS MOTOR EQUATION

Many real-life physical problems such as operation of synchronous motors, automatic frequency control, forced pendulum with constant torque, are described by the equation

$$\ddot{x} + a\dot{x} + b\sin x - c = 0, \quad a > 0, 0 < b < c, x(t) \in \mathbb{R}^1, \quad (10)$$

which was first studied by F. Tricomi in (Tricomi, 1933), also see (Andronov et al., 1966) and (Leonov et al., 1992). We are interested in checking the asymptotic stability of the equilibrium point $x = x^* = \arcsin(c/b)$, $y = \dot{x} = 0$. Traditionally, to analyze asymptotic stability, the full energy of the system is considered:

$$V = y^2/2 + b(\cos x^* - \cos x) - c(x - x^*).$$

For this function we have $\dot{V} = -W = -ay^2 \leq 0$, and the proof is accomplished by using LaSalle's principle. Instead, let us introduce a different Lyapunov function of the form

$$V(x, y) = \frac{y^2}{2} + \frac{ap}{2}(x - x^*)^2 + py(x - x^*) + b(\cos x^* - \cos x) - c(x - x^*), \quad (11)$$

where p is a design parameter. It is seen that $V(x^*, 0) = 0$, and $V(x, y) > 0$ in a certain neighborhood of $(x^*, 0)$; however this function can have local minima (for a small), as well as $\dot{V}$ does not globally sign-preserving. Hence, the results on asymptotic stability are only local.

Theorem 6. Let $0 < p < a$, $|x(0) + x^*| < \pi$, and

$$V(x(0), y(0)) < h = V(\pi - x^*, -p(\pi - x^*)).$$

Then system (10) is locally asymptotically stable:

$$x(t) \rightarrow x^*, \quad y(t) \rightarrow 0.$$

Proof. First, computing the derivatives we obtain

$$\dot{V} = -(a - p)y^2 - bp(x - x^*)(\sin x - \sin x^*)$$

and

$$W = (a - p)y^2 + bp(x - x^*)(\sin x - \sin x^*).$$

We immediately conclude that the conditions $0 < p < a$ and $|x + x^*| < \pi$ imply $W \geq 0$ and $W > 0$ for $(x, y) \neq (x^*, 0)$.

Second, we show that $V \geq 0$ for

$$(x, y) \in Q = \{x: |x + x^*| < \pi\}.$$

Denote $v(x) = \min_y V(x, y)$, then

$$y(x) = \arg \min_y V(x, y) = -p(x - x^*),$$

$$v(x) = \frac{(a - p)p}{2}(x - x^*)^2 + b(\cos x^* - \cos x) - c(x - x^*).$$

Hence,

$$v'(x) = (a - p)p(x - x^*) + b(\sin x - \sin x^*),$$

and $v(x) > 0$ for $|x + x^*| < \pi$, $(x, y) \neq (x^*, y)$. We conclude that $V(x, y) > 0$ if $|x + x^*| < \pi$, $(x, y) \neq (x^*, y)$; moreover the set

$$D = \{(x, y): V(x, y) < h, |x + x^*| < \pi\}$$

is bounded and lies in the interior of the set Q . Statement 3 of Theorem 1 now applies, which completes the proof.

Varying the parameter p we can change the domain of attraction D of the equilibrium point. The true basin of attraction is larger than its estimate D ; there are ways to extend it, because $W \geq 0$ is a less restrictive constraint than $(x, y) \in Q$.

The figure below depicts the trajectories of (10) for $a = 1$, $b = 0.8$, and $c = 0.4$, together with the domains D for various values of p .

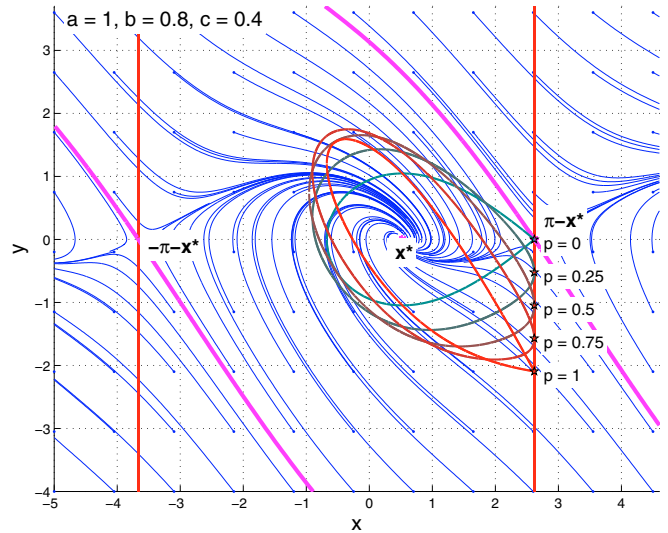


Fig. 1. Trajectories of system (10) with various initial conditions and the domains D for various values of p .

The basin of attraction in this example is defined by two separatrices (the two lines plotted in magenta); it is seen to be unbounded, meaning our approximations of the attraction domain are highly conservative for all p . However, for smaller values of the damping parameter a , the basin of attraction is semibounded (it belongs to the set $y \leq c$) and approximations are less conservative.

We do not consider the global behavior of the trajectories but rather focus just on the local asymptotic stability of the equilibrium point $(x^*, 0)$. It was F. Tricomi, see (Tricomi, 1933), who discovered two types of trajectories — rotatory motion and those convergent to equilibria points (there are countable amount of them); in some

cases (large damping) rotatory motion is lacking (as in the example under consideration). More details can be found in (Andronov et al., 1966; Leonov et al., 1992).

Such an approach can be generalized toward the multi-dimensional case for the following swing equation, which represents a dynamic model of a power system, see (Pai, 1989) and (Machowski et al., 2012) and a very recent publication (Weitenberg et al., 2016):

$$\ddot{x}_k + a_k \dot{x}_k + \sum_j b_{jk} \sin(x_k - x_j) - c_k = 0, \quad x_k(t) \in \mathbb{R}^1. \quad (12)$$

Important results in this direction have been recently obtained in (Vu and Turitsyn, 2016). They are based on the representation of (12) in the state-space form as

$$\dot{x} = Ax + B\varphi(Cx), \quad (13)$$

where A, B , and C are matrices and $\varphi(z)$ is a vector non-linearity. Then the Lyapunov function is designed as

$$V(x) = (Px, x) + K\psi(Cx),$$

where the matrices P, K are to be chosen, and $\psi'(z) = \varphi(z)$. Linear matrix inequalities (LMI) in the matrix variables P, K , were obtained in (Vu and Turitsyn, 2016), and their solution defined the functions $V, \dot{V}$ with the desired properties. It is interesting to note that a similar construction of the Lyapunov function goes back to the pioneering work on the absolute stability (Lur'e and Postnikov, 1944); this construction was later developed in numerous papers. However standard approaches to absolute stability do not use the LMI technique; see (Kamenetskiy and Pyatnitskiy, 1987; Rapoport, 2005; Boyd et al., 1994) among exceptions. Moreover, they guarantee the global asymptotic stability, while the approach in (Vu and Turitsyn, 2016) provides a less restrictive assumptions sufficient for local asymptotic stability. Compare Theorem 6 above with local results only; global asymptotic stability for system (10) is lacking.

5. CONCLUSIONS

To summarize, the main contributions of the paper are as follows.

- We highlighted various links between optimization problems and asymptotic stability analysis of differential equations. They lead to new approaches and provide more flexibility in both fields of research.
- Versions of Lyapunov's theorems under nonstandard assumptions are presented. For instance, neither the uniqueness of the minimum point of the Lyapunov function is assumed, nor its existence. Moreover, we proved asymptotic stability and obtained non-exponential rates of convergence.
- Novel types of Lyapunov functions for the heavy-ball method in optimization are proposed. They facilitate the proof of convergence together with finding estimates of the rate of convergence; this task is beyond the capabilities of the standard techniques.
- A similar approach exhibits its advantages for the synchronous motor equation. A nonstandard Lyapunov function admits for new estimates of the basin of attraction of equilibria points.

However we restricted ourselves to simplest situations and examples. The following issues are worth studying in the future:

- Non-autonomous systems (the right-hand side of (1) depends on t).
- Discrete-time systems that describe iterative methods of minimization.
- We did not optimize the parameters in the equations or in the Lyapunov functions to achieve the fastest convergence; our goal was just to demonstrate how does the technique work.
- We did not discuss numerical implementations of the methods; for instance, we omitted practical aspects of continuous-time methods for optimization.
- We restricted our attention to unconstrained smooth optimization; problems with non-smooth objective functions or constrained optimization problems deserve special analysis.
- We focused our attention exceptionally on the analysis of asymptotic stability; however, of a primary interest would be to consider the design problems and synthesize a feedback that leads to largest possible estimates of the domain of attraction of the closed-loop system.

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